

VuTienDat

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2025-11-23

```
# Câu 1:  
#install.packages("gap")  
library(gap)
```

```
## Warning: package 'gap' was built under R version 4.4.3
```

```
## Loading required package: gap.datasets
```

```
## gap version 1.6
```

```
library(MASS)  
data(hla)  
#str(hla)  
names(hla)
```

```
## [1] "id"      "y"      "DQR.a1" "DQR.a2" "DQA.a1" "DQA.a2" "DQB.a1" "DQB.a2"
```

```
#y la bien phan nhom  
#"DQR.a1" "DQR.a2" "DQA.a1" "DQA.a2" "DQB.a1" "DQB.a2" laf 6 dau hieu HLA
```

```
hla <- hla[,-1]  
hla$y <- factor(hla$y)  
  
model_lda <- lda(y ~ ., data = hla)  
model_lda
```

```
## Call:  
## lda(y ~ ., data = hla)  
##  
## Prior probabilities of groups:  
##      0      1  
## 0.6531365 0.3468635  
##  
## Group means:  
##      DQR.a1  DQR.a2  DQA.a1  DQA.a2  DQB.a1  DQB.a2  
## 0 9.350282 11.77401 3.553672 5.067797 5.327684 5.180791  
## 1 8.542553 14.24468 4.340426 4.223404 5.414894 6.712766  
##  
## Coefficients of linear discriminants:
```

```
##                LD1
## DQR.a1 -0.07709084
## DQR.a2  0.02727541
## DQA.a1  0.36753467
## DQA.a2 -0.08448677
## DQB.a1  0.14728570
## DQB.a2  0.13073260
```

```
#Dự đoán và kiểm tra độ chính xác
pred <- predict(model_lda)
#pred
table(Predict = pred$class, Real = hla$y)
```

```
##          Real
## Predict   0   1
##          0 156  67
##          1  21  27
```

```
#?mean
mean(pred$class == hla$y) #tạo ra một vector gồm true/false
```

```
## [1] 0.6752768
```

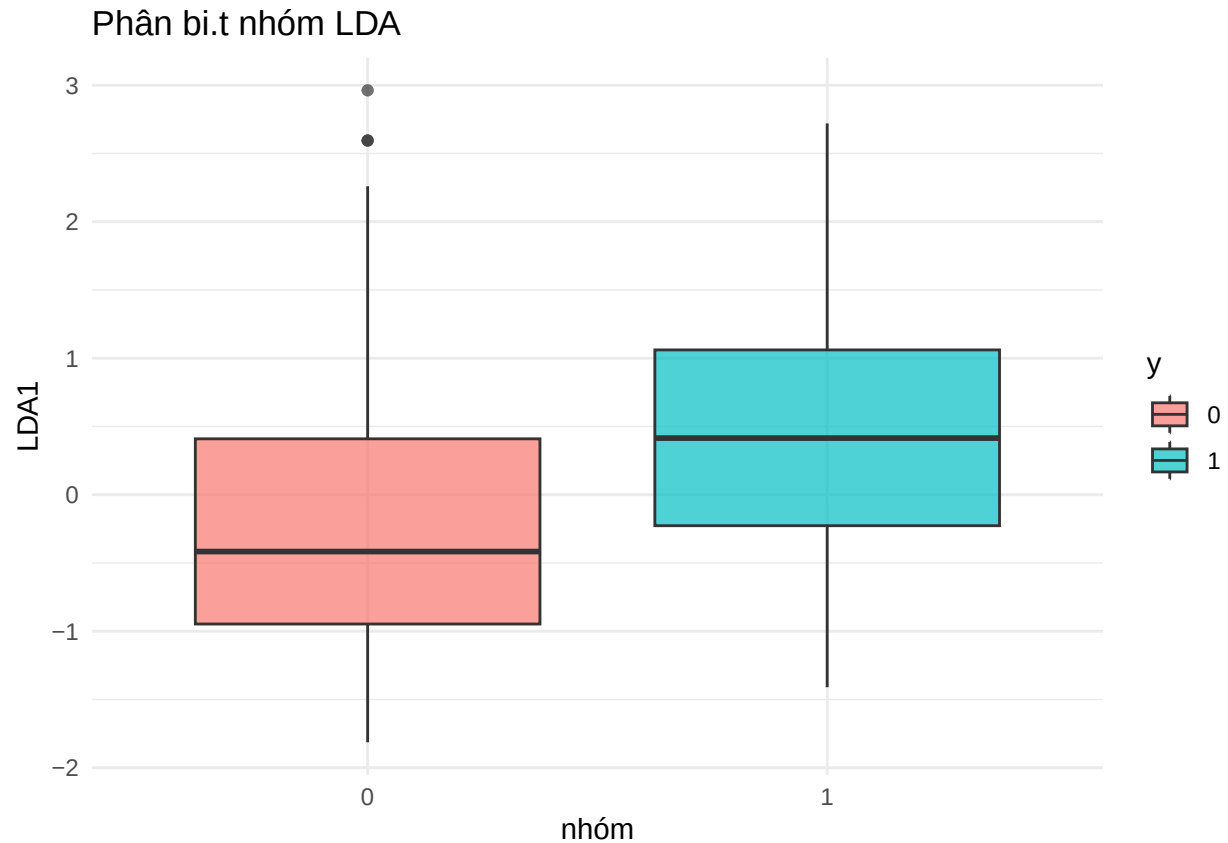
```
#=0.6752768 => mô hình dự đoán đúng 67,5% trong mẫu dữ liệu
```

```
#VẼ BIỂU ĐỒ LDA1
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
#dim(pred$x) #kiểm tra số trục phân biệt tối đa
lda_df <- data.frame(
  LDA1 = pred$x[,1],
  y = hla$y
)
```

```
ggplot(lda_df, aes(y, LDA1, fill = y)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Phân biệt nhóm LDA",
       x = "nhóm",
       y = "LDA1") +
  theme_minimal()
```



Câu 2:

```
library(MASS)
data(iris)
model_lda <- lda(Species~., data = iris)
model_lda
```

```
## Call:
## lda(Species ~ ., data = iris)
##
## Prior probabilities of groups:
##      setosa versicolor virginica
## 0.3333333 0.3333333 0.3333333
##
## Group means:
##      Sepal.Length Sepal.Width Petal.Length Petal.Width
## setosa           5.006      3.428      1.462      0.246
## versicolor       5.936      2.770      4.260      1.326
## virginica        6.588      2.974      5.552      2.026
##
## Coefficients of linear discriminants:
##      LD1      LD2
## Sepal.Length 0.8293776 -0.02410215
## Sepal.Width  1.5344731 -2.16452123
## Petal.Length -2.2012117 0.93192121
## Petal.Width  -2.8104603 -2.83918785
```

```
##
## Proportion of trace:
##      LD1      LD2
## 0.9912 0.0088
```

```
pred <- predict(model_lda)

table(Predict = pred$class, Real = iris$Species)
```

```
##           Real
## Predict      setosa versicolor virginica
##   setosa         50          0          0
##   versicolor      0          48          1
##   virginica       0           2         49
```

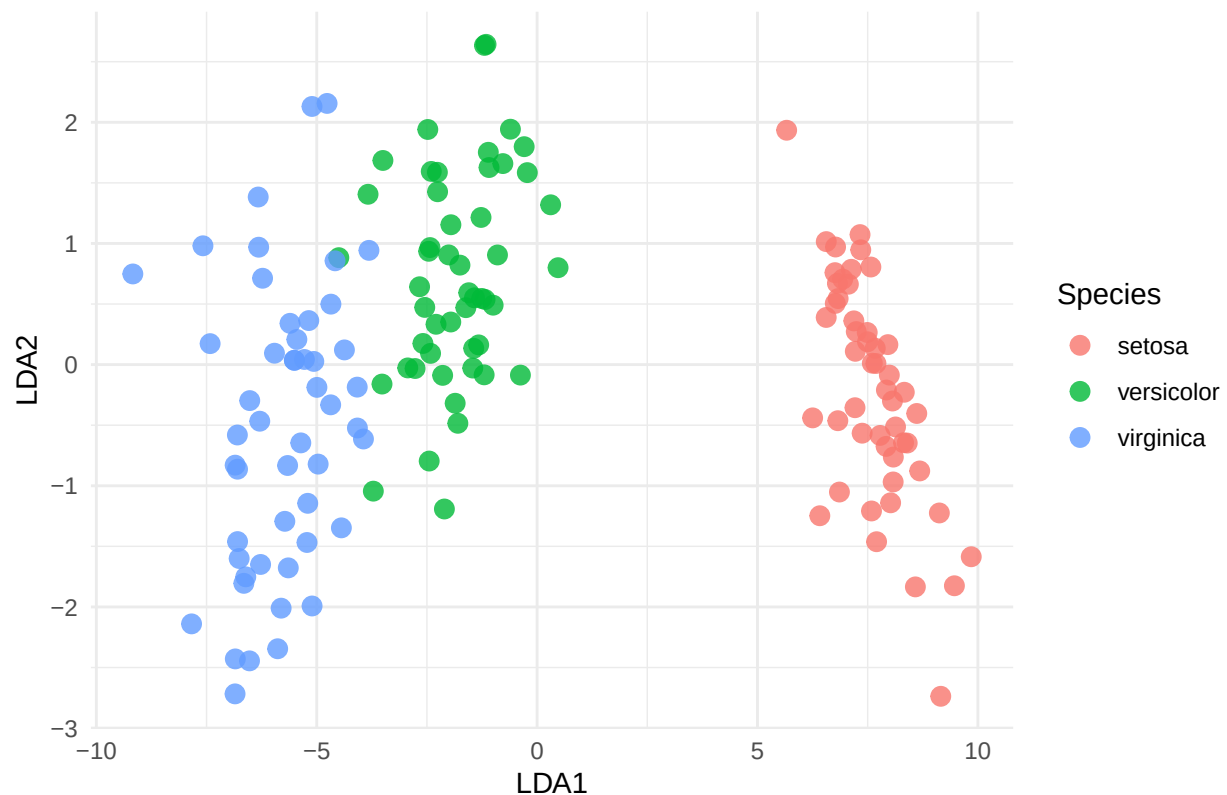
```
mean(pred$class == iris$Species)
```

```
## [1] 0.98
```

```
lda_df = data.frame(
  LDA1 = pred$x[,1],
  LDA2 = pred$x[,2],
  Species = iris$Species
)

library(ggplot2)
ggplot(lda_df, aes(LDA1, LDA2, color = Species)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "Đồ thị phân biệt LDA",
       x = "LDA1",
       y = "LDA2") +
  theme_minimal()
```

Đ. th. phân bi.t LDA



```
#HỆ số tải
model_lda$scaling
```

```
##              LD1          LD2
## Sepal.Length  0.8293776 -0.02410215
## Sepal.Width   1.5344731 -2.16452123
## Petal.Length -2.2012117  0.93192121
## Petal.Width  -2.8104603 -2.83918785
```

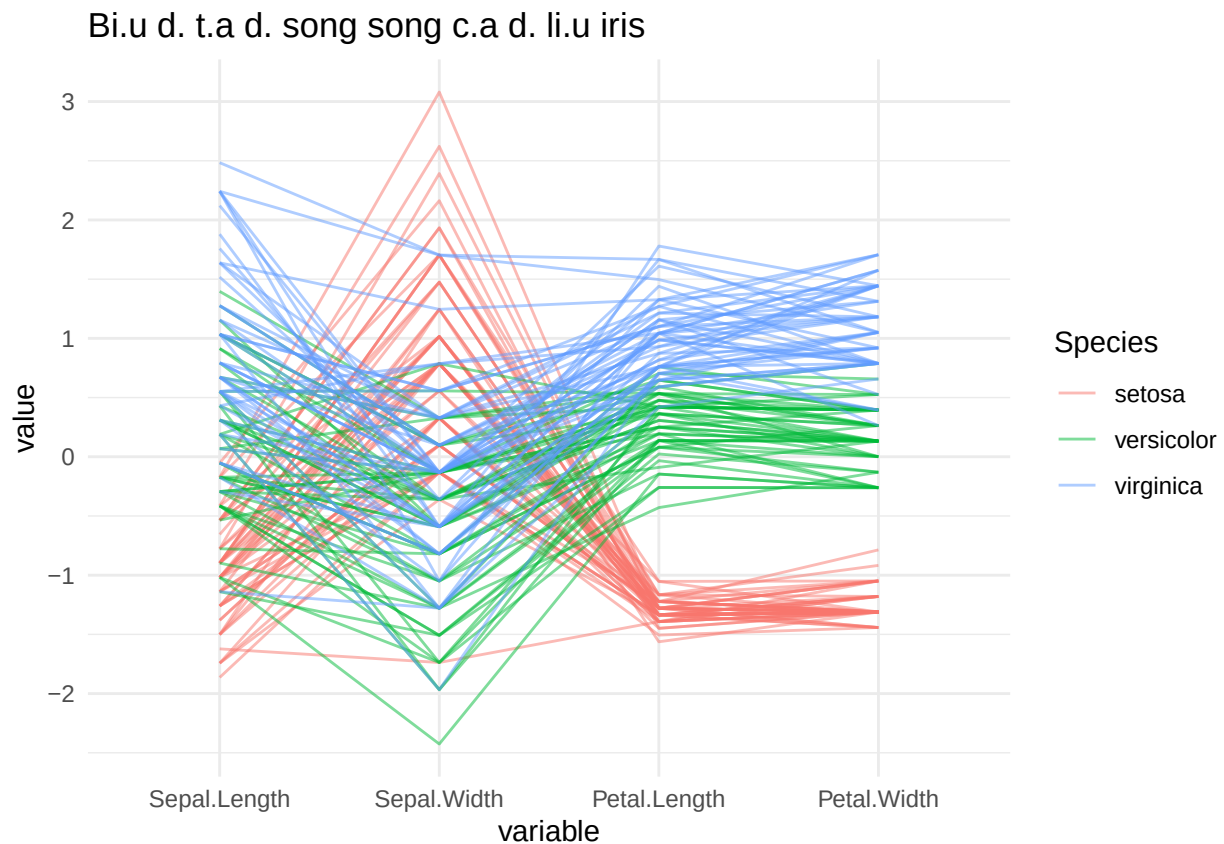
```
#LDA1 phân biệt setosa với 2 loài còn lại
#LDA1 chủ yếu dựa vào kích thước cánh hoa
#LDA2 phân biệt versicolor và virginica
#LDA2 chủ yếu dựa vào bề rộng đài hoa
```

```
#install.packages("GGally")
#Đối chiếu với biểu đồ tọa độ song song
library(GGally)
```

```
## Warning: package 'GGally' was built under R version 4.4.3
```

```
ggparcoord(data = iris,
  columns = 1:4,
  groupColumn = 5,
  scale = "std",
```

```
alphaLines = 0.5) +
theme_minimal() +
labs(title = "Biểu đồ tọa độ song song của dữ liệu iris")
```



```
# Câu 3:
# install.packages("kernlab")
# install.packages("fastDummies")

library(kernlab)
```

```
##
## Attaching package: 'kernlab'

## The following object is masked from 'package:ggplot2':
##
## alpha
```

```
library(MASS)
library(fastDummies)
```

```
## Warning: package 'fastDummies' was built under R version 4.4.3
```

```

# Load dữ liệu
data(promotergene)

# Biến Y
y <- promotergene$Class

# Biến X là 57 cột giá trị a, c, g, t
X <- promotergene[, -1]

# One-hot encoding cho tất cả các biến base
X_dummy <- fastDummies::dummy_cols(
  X,
  remove_first_dummy = TRUE,          # bỏ bớt 1 biến để tránh đa cộng tuyến
  remove_selected_columns = TRUE      # không giữ lại cột ký tự ban đầu
)

# Kiểm tra dữ liệu sau khi mã hóa
str(X_dummy)

```

```

## 'data.frame':   106 obs. of  171 variables:
## $ V2_c : int  0 0 1 0 0 0 1 0 1 0 ...
## $ V2_g : int  1 0 0 0 0 0 0 0 0 0 ...
## $ V2_t : int  0 0 0 1 1 0 0 1 0 1 ...
## $ V3_c : int  1 0 1 1 0 0 0 0 0 0 ...
## $ V3_g : int  0 0 0 0 0 0 0 0 0 1 ...
## $ V3_t : int  0 1 0 0 0 1 0 1 1 0 ...
## $ V4_c : int  1 0 0 0 0 0 0 0 0 0 ...
## $ V4_g : int  0 1 1 0 1 0 0 0 1 0 ...
## $ V4_t : int  0 0 0 1 0 0 0 0 0 1 ...
## $ V5_c : int  0 1 0 1 0 0 0 0 1 0 ...
## $ V5_g : int  0 0 0 0 1 0 1 1 0 0 ...
## $ V5_t : int  1 0 1 0 0 1 0 0 0 0 ...
## $ V6_c : int  0 0 0 0 1 0 0 1 0 0 ...
## $ V6_g : int  0 0 0 0 0 0 1 0 0 0 ...
## $ V6_t : int  1 0 1 0 0 0 0 0 0 0 ...
## $ V7_c : int  1 0 0 0 0 0 0 0 0 0 ...
## $ V7_g : int  0 0 0 0 0 0 0 1 0 0 ...
## $ V7_t : int  0 0 1 0 0 0 1 0 0 0 ...
## $ V8_c : int  0 0 0 1 1 0 0 0 0 1 ...
## $ V8_g : int  0 0 0 0 0 0 0 1 0 0 ...
## $ V8_t : int  1 1 0 0 0 0 0 0 1 0 ...
## $ V9_c : int  1 0 0 0 1 0 0 0 0 0 ...
## $ V9_g : int  0 0 0 1 0 0 1 0 0 0 ...
## $ V9_t : int  0 1 1 0 0 0 0 0 1 1 ...
## $ V10_c: int  1 0 0 0 1 0 0 0 0 0 ...
## $ V10_g: int  0 0 0 0 0 0 0 0 0 0 ...
## $ V10_t: int  0 1 1 1 0 0 0 1 1 0 ...
## $ V11_c: int  0 0 0 0 1 0 0 1 0 0 ...
## $ V11_g: int  0 0 0 0 0 0 0 0 0 0 ...
## $ V11_t: int  0 1 1 0 0 0 0 0 1 0 ...
## $ V12_c: int  0 0 0 0 0 0 0 1 0 0 ...
## $ V12_g: int  0 0 0 0 0 1 0 0 0 0 ...
## $ V12_t: int  0 1 1 0 0 0 1 0 1 1 ...

```

```

## $ V13_c: int 0 0 0 1 0 0 0 0 1 0 ...
## $ V13_g: int 0 0 0 0 1 0 1 0 0 1 ...
## $ V13_t: int 0 1 1 0 0 1 0 1 0 0 ...
## $ V14_c: int 0 0 0 0 0 0 1 0 0 1 ...
## $ V14_g: int 0 0 0 0 1 0 0 0 0 0 ...
## $ V14_t: int 0 0 1 0 0 1 0 0 1 0 ...
## $ V15_c: int 1 0 1 1 1 1 0 1 0 1 ...
## $ V15_g: int 0 1 0 0 0 0 0 0 0 0 ...
## $ V15_t: int 0 0 0 0 0 0 1 0 0 0 ...
## $ V16_c: int 0 0 0 0 0 0 0 1 0 0 ...
## $ V16_g: int 1 0 0 0 0 0 0 0 0 0 ...
## $ V16_t: int 0 1 1 1 1 1 1 0 1 1 ...
## $ V17_c: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V17_g: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V17_t: int 1 1 0 1 1 1 1 1 1 1 ...
## $ V18_c: int 0 0 1 0 0 0 0 0 0 0 ...
## $ V18_g: int 1 1 0 0 0 1 1 1 1 0 ...
## $ V18_t: int 0 0 0 1 1 0 0 0 0 1 ...
## $ V19_c: int 0 1 1 0 0 1 1 0 1 0 ...
## $ V19_g: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V19_t: int 1 0 0 0 0 0 0 0 0 0 ...
## $ V20_c: int 0 0 1 1 1 0 1 1 0 1 ...
## $ V20_g: int 0 0 0 0 0 0 0 0 1 0 ...
## $ V20_t: int 1 0 0 0 0 1 0 0 0 0 ...
## $ V21_c: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V21_g: int 0 0 0 0 0 0 0 1 1 1 ...
## $ V21_t: int 1 1 0 0 0 1 1 0 0 0 ...
## $ V22_c: int 0 0 0 0 1 0 0 1 1 0 ...
## $ V22_g: int 0 1 0 1 0 0 0 0 0 0 ...
## $ V22_t: int 1 0 1 0 0 1 1 0 0 1 ...
## $ V23_c: int 0 0 0 1 0 1 0 0 1 0 ...
## $ V23_g: int 0 0 0 0 0 0 1 0 0 1 ...
## $ V23_t: int 1 0 0 0 1 0 0 1 0 0 ...
## $ V24_c: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V24_g: int 0 0 0 1 0 0 0 0 0 1 ...
## $ V24_t: int 1 0 1 0 1 1 1 1 1 0 ...
## $ V25_c: int 0 1 1 0 0 0 1 0 0 0 ...
## $ V25_g: int 1 0 0 1 0 0 0 0 1 1 ...
## $ V25_t: int 0 0 0 0 1 0 0 1 0 0 ...
## $ V26_c: int 0 0 1 1 0 0 0 0 1 1 ...
## $ V26_g: int 0 0 0 0 0 0 1 0 0 0 ...
## $ V26_t: int 1 1 0 0 0 0 0 1 0 0 ...
## $ V27_c: int 0 1 0 0 0 1 0 0 0 0 ...
## $ V27_g: int 0 0 0 1 0 0 1 0 1 1 ...
## $ V27_t: int 1 0 1 0 1 0 0 1 0 0 ...
## $ V28_c: int 0 0 0 1 0 0 1 0 0 0 ...
## $ V28_g: int 1 1 0 0 1 1 0 0 1 1 ...
## $ V28_t: int 0 0 1 0 0 0 0 0 0 0 ...
## $ V29_c: int 0 1 0 0 1 0 1 0 0 0 ...
## $ V29_g: int 0 0 1 1 0 0 0 0 0 0 ...
## $ V29_t: int 1 0 0 0 0 1 0 1 0 1 ...
## $ V30_c: int 0 0 0 0 0 0 0 1 0 0 ...
## $ V30_g: int 0 0 0 0 0 1 0 0 1 1 ...
## $ V30_t: int 1 0 0 1 1 0 1 0 0 0 ...

```

```
## $ V31_c: int 0 0 0 1 0 0 0 0 0 0 ...
## $ V31_g: int 0 0 0 0 0 0 1 1 0 0 ...
## $ V31_t: int 0 1 0 0 1 0 0 0 0 0 ...
## $ V32_c: int 0 0 0 0 1 0 0 1 0 0 ...
## $ V32_g: int 0 1 1 0 0 0 0 0 0 0 ...
## $ V32_t: int 0 0 0 0 0 0 0 0 0 1 ...
## $ V33_c: int 0 0 1 0 1 0 0 0 1 0 ...
## $ V33_g: int 0 0 0 0 0 0 0 0 0 0 ...
## $ V33_t: int 1 1 0 1 0 0 1 0 0 1 ...
## $ V34_c: int 0 1 0 0 0 0 0 0 0 0 ...
## $ V34_g: int 0 0 1 0 1 1 0 0 0 0 ...
## $ V34_t: int 1 0 0 1 0 0 1 0 1 1 ...
## [list output truncated]
```

```
# LDA
```

```
model <- lda(y ~ ., data = X_dummy)
```

```
## Warning in lda.default(x, grouping, ...): variables are collinear
```

```
model
```

```
## Call:
```

```
## lda(y ~ ., data = X_dummy)
```

```
##
```

```
## Prior probabilities of groups:
```

```
## + -
```

```
## 0.5 0.5
```

```
##
```

```
## Group means:
```

```
##      V2_c      V2_g      V2_t      V3_c      V3_g      V3_t      V4_c
## + 0.2264151 0.1509434 0.3584906 0.2830189 0.1320755 0.2641509 0.1886792
## - 0.2830189 0.1320755 0.3584906 0.1320755 0.3207547 0.2264151 0.2075472
##      V4_g      V4_t      V5_c      V5_g      V5_t      V6_c      V6_g
## + 0.2830189 0.2641509 0.2641509 0.2641509 0.2641509 0.1886792 0.1886792
## - 0.2452830 0.2452830 0.3018868 0.2641509 0.2264151 0.1698113 0.3584906
##      V6_t      V7_c      V7_g      V7_t      V8_c      V8_g      V8_t
## + 0.2075472 0.1132075 0.09433962 0.1886792 0.1698113 0.1132075 0.2264151
## - 0.2075472 0.2264151 0.32075472 0.2641509 0.2264151 0.2075472 0.3396226
##      V9_c      V9_g      V9_t      V10_c      V10_g      V10_t      V11_c
## + 0.09433962 0.1320755 0.3962264 0.1698113 0.09433962 0.4150943 0.1320755
## - 0.28301887 0.2452830 0.2075472 0.2452830 0.26415094 0.1886792 0.2830189
##      V11_g      V11_t      V12_c      V12_g      V12_t      V13_c      V13_g
## + 0.1320755 0.2452830 0.1132075 0.1320755 0.3396226 0.3018868 0.1886792
## - 0.2452830 0.2830189 0.3018868 0.1509434 0.2452830 0.2830189 0.2830189
##      V13_t      V14_c      V14_g      V14_t      V15_c      V15_g      V15_t
## + 0.3396226 0.09433962 0.2641509 0.3962264 0.4339623 0.2641509 0.09433962
## - 0.2075472 0.16981132 0.2830189 0.2452830 0.2830189 0.1698113 0.30188679
##      V16_c      V16_g      V16_t      V17_c      V17_g      V17_t      V18_c
## + 0.03773585 0.1320755 0.8113208 0.09433962 0.03773585 0.8113208 0.0754717
## - 0.20754717 0.1698113 0.2075472 0.35849057 0.16981132 0.2075472 0.1886792
##      V18_g      V18_t      V19_c      V19_g      V19_t      V20_c      V20_g
## + 0.7924528 0.1320755 0.2452830 0.01886792 0.2075472 0.5660377 0.0754717
## - 0.2075472 0.3207547 0.3018868 0.33962264 0.1320755 0.2641509 0.1509434
```

```

##      V20_t      V21_c      V21_g      V21_t      V22_c      V22_g      V22_t
## + 0.1320755 0.09433962 0.2075472 0.2830189 0.2452830 0.1320755 0.3584906
## - 0.3018868 0.35849057 0.3018868 0.1698113 0.1698113 0.2830189 0.2830189
##      V23_c      V23_g      V23_t      V24_c      V24_g      V24_t      V25_c
## + 0.2075472 0.2264151 0.245283 0.2452830 0.2830189 0.3207547 0.2264151
## - 0.1698113 0.3207547 0.245283 0.2641509 0.2830189 0.1886792 0.1698113
##      V25_g      V25_t      V26_c      V26_g      V26_t      V27_c      V27_g
## + 0.2264151 0.3584906 0.1698113 0.2264151 0.3396226 0.09433962 0.2641509
## - 0.3207547 0.2452830 0.2264151 0.2264151 0.2641509 0.16981132 0.3773585
##      V27_t      V28_c      V28_g      V28_t      V29_c      V29_g      V29_t
## + 0.3584906 0.3018868 0.3018868 0.2075472 0.2830189 0.2075472 0.3018868
## - 0.2075472 0.1698113 0.2452830 0.2452830 0.2830189 0.3018868 0.3018868
##      V30_c      V30_g      V30_t      V31_c      V31_g      V31_t      V32_c
## + 0.2075472 0.3207547 0.3396226 0.1320755 0.3018868 0.1698113 0.1509434
## - 0.1698113 0.3018868 0.3207547 0.3584906 0.1886792 0.1886792 0.1698113
##      V32_g      V32_t      V33_c      V33_g      V33_t      V34_c      V34_g
## + 0.1320755 0.2264151 0.2075472 0.1886792 0.3207547 0.1698113 0.1320755
## - 0.2641509 0.3584906 0.2075472 0.3396226 0.1509434 0.2830189 0.1886792
##      V34_t      V35_c      V35_g      V35_t      V36_c      V36_g      V36_t
## + 0.4905660 0.2452830 0.2075472 0.3584906 0.3018868 0.2641509 0.1886792
## - 0.2830189 0.1698113 0.2830189 0.3773585 0.2075472 0.2075472 0.3207547
##      V37_c      V37_g      V37_t      V38_c      V38_g      V38_t      V39_c
## + 0.2452830 0.2264151 0.3018868 0.1698113 0.245283 0.4716981 0.1132075
## - 0.2075472 0.2641509 0.3396226 0.1509434 0.245283 0.3396226 0.2830189
##      V39_g      V39_t      V40_c      V40_g      V40_t      V41_c      V41_g
## + 0.1698113 0.3773585 0.0754717 0.01886792 0.4528302 0.0754717 0.09433962
## - 0.3207547 0.2264151 0.3018868 0.30188679 0.2075472 0.1698113 0.30188679
##      V41_t      V42_c      V42_g      V42_t      V43_c      V43_g      V43_t
## + 0.2452830 0.1320755 0.1698113 0.1320755 0.1698113 0.1886792 0.3962264
## - 0.2641509 0.2452830 0.1698113 0.3773585 0.3396226 0.3207547 0.2264151
##      V44_c      V44_g      V44_t      V45_c      V45_g      V45_t      V46_c
## + 0.09433962 0.2641509 0.4716981 0.3207547 0.1509434 0.3584906 0.18867925
## - 0.22641509 0.2075472 0.3207547 0.2641509 0.2075472 0.3207547 0.09433962
##      V46_g      V46_t      V47_c      V47_g      V47_t      V48_c      V48_g
## + 0.2830189 0.2264151 0.4528302 0.2075472 0.1886792 0.3396226 0.1886792
## - 0.3584906 0.3207547 0.2075472 0.2452830 0.2264151 0.2452830 0.2641509
##      V48_t      V49_c      V49_g      V49_t      V50_c      V50_g      V50_t
## + 0.2452830 0.3773585 0.1698113 0.1886792 0.4716981 0.2830189 0.0754717
## - 0.2075472 0.2264151 0.2452830 0.3207547 0.2075472 0.2075472 0.3207547
##      V51_c      V51_g      V51_t      V52_c      V52_g      V52_t      V53_c
## + 0.4528302 0.1509434 0.1698113 0.3773585 0.1698113 0.1698113 0.3018868
## - 0.3396226 0.1886792 0.2452830 0.2075472 0.2830189 0.2641509 0.3018868
##      V53_g      V53_t      V54_c      V54_g      V54_t      V55_c      V55_g
## + 0.05660377 0.3396226 0.1509434 0.2075472 0.3962264 0.2830189 0.2830189
## - 0.20754717 0.2830189 0.2452830 0.2641509 0.2641509 0.3207547 0.1320755
##      V55_t      V56_c      V56_g      V56_t      V57_c      V57_g      V57_t
## + 0.2264151 0.3207547 0.2641509 0.1698113 0.2830189 0.2452830 0.2264151
## - 0.3396226 0.2264151 0.2641509 0.2641509 0.2641509 0.2075472 0.3207547
##      V58_c      V58_g      V58_t
## + 0.1509434 0.2452830 0.3962264
## - 0.1698113 0.2830189 0.2452830
##
## Coefficients of linear discriminants:
##      LD1

```

```
## V2_c -0.594499103
## V2_g -0.205336037
## V2_t 0.147111453
## V3_c 0.431191347
## V3_g 0.599447110
## V3_t 0.297409997
## V4_c -0.554096833
## V4_g -0.188116830
## V4_t -0.428115239
## V5_c 0.119744417
## V5_g -0.928133284
## V5_t -0.663729067
## V6_c 0.884011389
## V6_g 1.039460337
## V6_t 0.820381045
## V7_c -0.080523328
## V7_g 0.208462223
## V7_t -0.284283417
## V8_c 0.584913582
## V8_g 0.652319345
## V8_t 1.249423192
## V9_c -0.095060063
## V9_g 0.067348780
## V9_t -0.348784679
## V10_c -0.117412961
## V10_g -0.103574864
## V10_t -0.373752289
## V11_c 0.435822261
## V11_g 0.828704448
## V11_t -0.401519115
## V12_c 0.254788681
## V12_g -0.007565121
## V12_t 1.411975030
## V13_c 0.433121048
## V13_g -0.044425339
## V13_t -0.034677318
## V14_c -1.356662812
## V14_g 0.652470948
## V14_t -1.246311949
## V15_c -0.393732894
## V15_g 0.391808908
## V15_t 1.140603028
## V16_c 0.581686189
## V16_g -0.222864383
## V16_t -0.977249105
## V17_c -0.479168343
## V17_g 1.462342075
## V17_t -0.286570852
## V18_c 0.406185101
## V18_g -1.043782251
## V18_t -0.035012608
## V19_c -0.248129795
## V19_g -0.009522576
## V19_t 0.074631042
```

```
## V20_c 0.072373914
## V20_g 0.370154770
## V20_t 0.397526092
## V21_c 0.537149908
## V21_g 0.775078061
## V21_t 0.226526427
## V22_c -0.351208104
## V22_g -0.348205970
## V22_t 0.485164808
## V23_c 0.711442275
## V23_g 0.668082599
## V23_t -0.610720892
## V24_c -0.179003062
## V24_g -0.035741683
## V24_t -0.270269681
## V25_c -0.052583540
## V25_g -0.349557754
## V25_t -0.406663184
## V26_c -0.916542173
## V26_g -0.396767609
## V26_t -0.806439286
## V27_c -0.016539954
## V27_g -0.089259364
## V27_t -0.491582292
## V28_c 0.328523653
## V28_g -0.527482819
## V28_t 0.362587332
## V29_c -0.189255476
## V29_g 0.076908866
## V29_t -0.212157958
## V30_c 0.138700668
## V30_g -0.388145560
## V30_t -0.412826657
## V31_c 0.991677544
## V31_g 0.128765790
## V31_t -0.314853823
## V32_c -0.325229414
## V32_g 1.265039786
## V32_t -0.796638427
## V33_c 0.456359985
## V33_g 0.428370135
## V33_t 0.198925622
## V34_c -0.844585455
## V34_g 0.609826216
## V34_t -0.405807664
## V35_c -0.232470324
## V35_g -0.006419912
## V35_t 0.341406502
## V36_c -0.073886949
## V36_g -0.170654649
## V36_t 0.620209335
## V37_c 0.192121707
## V37_g 0.708666360
## V37_t 0.201673484
```

V38_c -0.264890599
V38_g -0.008321200
V38_t 0.003908039
V39_c 0.682434159
V39_g 0.257060180
V39_t -0.158238726
V40_c 0.064794022
V40_g 0.448348892
V40_t -0.109869549
V41_c 0.696184777
V41_g 0.231078293
V41_t -0.177347741
V42_c -0.251094789
V42_g -0.135408446
V42_t 0.395165203
V43_c 0.104689074
V43_g 0.444566113
V43_t 0.639559918
V44_c 0.536270330
V44_g -0.312736405
V44_t -0.110835349
V45_c 0.016311499
V45_g -1.001415030
V45_t -0.475942332
V46_c 0.180506344
V46_g 0.293083860
V46_t 0.291023988
V47_c -0.268078279
V47_g 0.587987778
V47_t -0.075694799
V48_c 0.226887016
V48_g -0.229120610
V48_t -0.594718487
V49_c -0.862246029
V49_g 0.167610325
V49_t -0.002637937
V50_c -0.161119446
V50_g 0.102870498
V50_t 1.486013256
V51_c -0.287536172
V51_g -0.110786488
V51_t -0.582257968
V52_c 0.015165631
V52_g 0.034443535
V52_t -0.161388089
V53_c -0.314141295
V53_g -0.464906803
V53_t -0.259663090
V54_c -0.196862051
V54_g 0.294710563
V54_t -0.022726873
V55_c -0.120321753
V55_g -0.804699068
V55_t 0.088658755

```
## V56_c -0.105284028
## V56_g  0.874449666
## V56_t  0.248106352
## V57_c  0.284913173
## V57_g -0.173058513
## V57_t -0.167483476
## V58_c -0.330533003
## V58_g -0.627140845
## V58_t -0.449624075
```

```
# Dự đoán
pred <- predict(model)$class
accuracy <- mean(pred == y)
accuracy
```

```
## [1] 1
```

```
lda_out <- predict(model)
```

```
# Câu 4:
library(kernlab)
data(spam)
head(spam)
```

```
##  make address  all num3d  our over remove internet order mail receive will
## 1 0.00 0.64 0.64 0 0.32 0.00 0.00 0.00 0.00 0.00 0.00 0.64
## 2 0.21 0.28 0.50 0 0.14 0.28 0.21 0.07 0.00 0.94 0.21 0.79
## 3 0.06 0.00 0.71 0 1.23 0.19 0.19 0.12 0.64 0.25 0.38 0.45
## 4 0.00 0.00 0.00 0 0.63 0.00 0.31 0.63 0.31 0.63 0.31 0.31
## 5 0.00 0.00 0.00 0 0.63 0.00 0.31 0.63 0.31 0.63 0.31 0.31
## 6 0.00 0.00 0.00 0 1.85 0.00 0.00 1.85 0.00 0.00 0.00 0.00
##  people report addresses free business email you credit your font num000
## 1 0.00 0.00 0.00 0.32 0.00 1.29 1.93 0.00 0.96 0 0.00
## 2 0.65 0.21 0.14 0.14 0.07 0.28 3.47 0.00 1.59 0 0.43
## 3 0.12 0.00 1.75 0.06 0.06 1.03 1.36 0.32 0.51 0 1.16
## 4 0.31 0.00 0.00 0.31 0.00 0.00 3.18 0.00 0.31 0 0.00
## 5 0.31 0.00 0.00 0.31 0.00 0.00 3.18 0.00 0.31 0 0.00
## 6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0 0.00
##  money hp hpl george num650 lab labs telnet num857 data num415 num85
## 1 0.00 0 0 0 0 0 0 0 0 0 0 0
## 2 0.43 0 0 0 0 0 0 0 0 0 0 0
## 3 0.06 0 0 0 0 0 0 0 0 0 0 0
## 4 0.00 0 0 0 0 0 0 0 0 0 0 0
## 5 0.00 0 0 0 0 0 0 0 0 0 0 0
## 6 0.00 0 0 0 0 0 0 0 0 0 0 0
##  technology num1999 parts pm direct cs meeting original project re edu
## 1 0 0.00 0 0 0.00 0 0 0.00 0 0.00 0 0.00 0.00
## 2 0 0.07 0 0 0.00 0 0 0.00 0 0.00 0 0.00 0.00
## 3 0 0.00 0 0 0.06 0 0 0.12 0 0.06 0.06
## 4 0 0.00 0 0 0.00 0 0 0.00 0 0.00 0.00
## 5 0 0.00 0 0 0.00 0 0 0.00 0 0.00 0.00
## 6 0 0.00 0 0 0.00 0 0 0.00 0 0.00 0.00
```

```
##      table conference charSemicolon charRoundbracket charSquarebracket
## 1      0          0          0.00          0.000          0
## 2      0          0          0.00          0.132          0
## 3      0          0          0.01          0.143          0
## 4      0          0          0.00          0.137          0
## 5      0          0          0.00          0.135          0
## 6      0          0          0.00          0.223          0
##      charExclamation charDollar charHash capitalAve capitalLong capitalTotal type
## 1          0.778      0.000      0.000      3.756          61          278 spam
## 2          0.372      0.180      0.048      5.114         101         1028 spam
## 3          0.276      0.184      0.010      9.821         485         2259 spam
## 4          0.137      0.000      0.000      3.537          40          191 spam
## 5          0.135      0.000      0.000      3.537          40          191 spam
## 6          0.000      0.000      0.000      3.000          15           54 spam
```

```
#LDA
```

```
lda_spam = lda(type ~ ., data = spam)
print(lda_spam)
```

```
## Call:
## lda(type ~ ., data = spam)
##
## Prior probabilities of groups:
##      nonspam      spam
## 0.6059552 0.3940448
##
## Group means:
##      make      address      all      num3d      our      over
## nonspam 0.0734792 0.2444656 0.2005811 0.0008859397 0.1810402 0.04454448
## spam    0.1523387 0.1646498 0.4037948 0.1646718147 0.5139548 0.17487590
##      remove      internet      order      mail      receive      will
## nonspam 0.00938307 0.03841463 0.03804878 0.1671700 0.0217109 0.5363235
## spam    0.27540541 0.20814120 0.17006067 0.3505074 0.1184335 0.5499724
##      people      report      addresses      free      business      email
## nonspam 0.06166428 0.04240316 0.008317791 0.0735868 0.04834648 0.09729197
## spam    0.14354661 0.08357419 0.112079426 0.5183618 0.28750689 0.31922780
##      you      credit      your      font      num000      money
## nonspam 1.270341 0.00757891 0.4387016 0.04522597 0.007087518 0.01713773
## spam    2.264539 0.20552124 1.3803696 0.23803640 0.247054606 0.21287921
##      hp      hpl      george      num650      lab      labs
## nonspam 0.89547346 0.431994261 1.265265423 0.19380560 0.1627941176 0.165853659
## spam    0.01747932 0.009172642 0.001549917 0.01879757 0.0006839493 0.005968009
##      telnet      num857      data      num415      num85      technology
## nonspam 0.106032999 0.0773063128 0.1509864 0.077786944 0.169454806 0.14167145
## spam    0.001274131 0.0005184777 0.0145615 0.001776062 0.006927744 0.02951462
##      num1999      parts      pm      direct      cs      meeting
## nonspam 0.19774390 0.018723099 0.12167862 0.08311693 0.0720265423 0.216807747
## spam    0.04346939 0.004710425 0.01242692 0.03671815 0.0000551572 0.002443464
##      original      project      re      edu      table      conference
## nonspam 0.070581062 0.126635581 0.4157604 0.28718436 0.008192253 0.051226686
## spam    0.008450083 0.006243795 0.1250910 0.01472697 0.001218974 0.002101489
##      charSemicolon charRoundbracket charSquarebracket charExclamation
## nonspam 0.05028085      0.1585782      0.022683644      0.1099835
## spam    0.02057308      0.1089702      0.008198566      0.5137126
```

```

##          charDollar  charHash capitalAve capitalLong capitalTotal
## nonspam 0.01164849 0.02171306  2.377301   18.21449   161.4709
## spam    0.17447821 0.07887700  9.519165  104.39327   470.6194
##
## Coefficients of linear discriminants:
##                               LD1
## make                          -0.2053433845
## address                       -0.0496520077
## all                           0.1618979041
## num3d                         0.0491205095
## our                           0.3470862316
## over                          0.4898352934
## remove                        0.8776953914
## internet                      0.3874021379
## order                         0.2987224576
## mail                          0.0621045827
## receive                       0.2343512301
## will                          -0.1148308781
## people                        0.0490659059
## report                        0.0200317976
## addresses                     0.0763541263
## free                          0.3093868784
## business                      0.2131611128
## email                         0.2283383726
## you                           0.0582547287
## credit                        0.2544047910
## your                          0.2171922810
## font                          0.1845200015
## num000                        0.7204900016
## money                          0.3746324145
## hp                            -0.0955222618
## hpl                           -0.0891499158
## george                        -0.0502923959
## num650                        0.0164352762
## lab                           -0.0307054282
## labs                          -0.2141201057
## telnet                        -0.0960127853
## num857                        0.0260981607
## data                          -0.1730466665
## num415                        0.2107954203
## num85                         -0.1284707897
## technology                    0.1091451554
## num1999                       -0.1368948926
## parts                         -0.2202625065
## pm                            -0.0814242179
## direct                        0.1680076688
## cs                            -0.0344749731
## meeting                       -0.1522032232
## original                      -0.2606575435
## project                       -0.1334635603
## re                            -0.1453064935
## edu                           -0.1558616802
## table                         -0.8044795076
## conference                     -0.2399825033

```

```
## charSemicolon      -0.5774724823
## charRoundbracket  -0.2471396406
## charSquarebracket -0.2433979819
## charExclamation    0.2804998615
## charDollar         0.9611097972
## charHash           0.1141464080
## capitalAve         0.0009590191
## capitalLong        0.0002751450
## capitalTotal       0.0003291749
```

```
#vecto trọng số
lda_spam$scaling
```

```
##                               LD1
## make                        -0.2053433845
## address                    -0.0496520077
## all                        0.1618979041
## num3d                      0.0491205095
## our                        0.3470862316
## over                       0.4898352934
## remove                    0.8776953914
## internet                  0.3874021379
## order                     0.2987224576
## mail                      0.0621045827
## receive                   0.2343512301
## will                      -0.1148308781
## people                    0.0490659059
## report                    0.0200317976
## addresses                 0.0763541263
## free                      0.3093868784
## business                  0.2131611128
## email                    0.2283383726
## you                       0.0582547287
## credit                   0.2544047910
## your                     0.2171922810
## font                     0.1845200015
## num000                   0.7204900016
## money                    0.3746324145
## hp                       -0.0955222618
## hpl                      -0.0891499158
## george                   -0.0502923959
## num650                   0.0164352762
## lab                      -0.0307054282
## labs                     -0.2141201057
## telnet                   -0.0960127853
## num857                   0.0260981607
## data                     -0.1730466665
## num415                   0.2107954203
## num85                    -0.1284707897
## technology               0.1091451554
## num1999                 -0.1368948926
## parts                    -0.2202625065
## pm                      -0.0814242179
```

```
## direct      0.1680076688
## cs          -0.0344749731
## meeting     -0.1522032232
## original    -0.2606575435
## project     -0.1334635603
## re          -0.1453064935
## edu         -0.1558616802
## table       -0.8044795076
## conference  -0.2399825033
## charSemicolon -0.5774724823
## charRoundbracket -0.2471396406
## charSquarebracket -0.2433979819
## charExclamation 0.2804998615
## charDollar    0.9611097972
## charHash      0.1141464080
## capitalAve    0.0009590191
## capitalLong   0.0002751450
## capitalTotal  0.0003291749
```

```
#
scores = predict(lda_spam)$x
plot(scores, col = "red", pch = 16, cex = .3)
```

